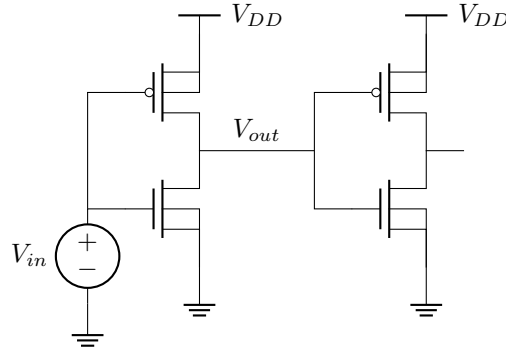


EC 5311 Digital IC Design: Assignment 4
CMOS inverter - transient characteristics and Ring oscillator

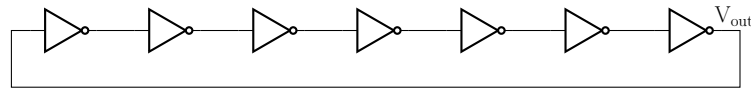
$$V_{Tn} = 0.7V, \quad \mu_n = 0.025\text{m}^2/\text{V} - \text{s}, \quad C_{oxn} = 8.34\text{fF}/\mu\text{m}^2, \quad v_{satn} = 8 \times 10^4\text{m/s}, \quad \lambda_n = 0.2/\text{V}$$

$$|V_{Tp}| = 0.7V, \quad \mu_p = 0.009\text{m}^2/\text{V} - \text{s}, \quad C_{oxp} = 8.16\text{fF}/\mu\text{m}^2, \quad v_{satp} = 3 \times 10^4\text{m/s}, \quad \lambda_p = 0.2/\text{V}$$

1. Consider the static CMOS inverter shown below. It drives another identical inverter. The input V_{in} is a pulse between 0 and V_{DD} , with a rise and fall time equal to 5ps and a pulse width of 250ps. The output is V_{out} .



- (a) Set $V_{DD} = 1.8V$. Assume that $L_n = L_p = 0.15\mu\text{m}$ and $W_n = 0.42\mu\text{m}$. Obtain the delay for $W_p = 0.42\mu\text{m}, 0.84\mu\text{m}, 1.26\mu\text{m}$.
 - (b) Set W_p so that delay is minimised. Plot the delay as a function of V_{DD} for $V_{DD} = 1V$ to $1.8V$ in steps of $0.1V$. How does it compare with analytical estimates?
 - (c) Plot the measured and estimated energy-delay product as a function of V_{DD} . What is the optimum V_{DD} ?
2. Using the CMOS inverter with the minimum delay from Question 1(a), construct a seven stage ring oscillator as shown below and find the oscillating frequency.



To ensure oscillation in the transient simulation, set the node $V_{out} = 0V$ initially using: `.ic v(Vout)=0`

- (a) Measure the oscillating frequency for $V_{DD} = 1.8V$.
- (b) Plot the oscillating frequency and time period as a function of V_{DD} for $V_{DD} = 1V$ to $1.8V$ in steps of $0.1V$.
- (c) Repeat the experiment (b) with nine inverter stages.